

Multimodal Data Fusion in IoMT For Early Disease Detection using Deep Learning with Generated Data

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Abstract—In healthcare, Electronic Health Records (EHRs) and continuous patient monitoring data are restricted by strict privacy regulations like GDPR in the European Union and HIPAA in the USA. To overcome this obstacle in medical AI development, we developed a rule-based synthetic data generation program that simulates a real-world Internet of Medical Things (IoMT) environment. The pipeline generated 500,000 synthetic patient records, including continuous vital signs, demographic history, and laboratory biomarkers across 24 distinct medical pathologies. To demonstrate the clinical representativeness of this dataset, we validated its statistical distributions against a real-world open-access Emergency Department admission dataset. Furthermore, we benchmarked an Early Fusion Multi-Layer Perceptron (MLP), a Contextual Fusion Tabular Transformer, and a tree-based baseline (XGBoost). Our results demonstrate that the synthetic data poses a challenging classification task. While models achieved a high ROC-AUC by effectively mapping multidimensional boundaries, deliberate overlapping clinical phenotypes mirrored real-world diagnostic ambiguity, limiting Macro F1-scores appropriately. Finally, SHapley Additive exPlanations (SHAP) confirmed that the algorithms learned genuine expert-driven physiological rules, suggesting this framework as a useful tool for privacy-preserving medical predictive modeling.

Index Terms—Synthetic Data Generation, Electronic Health Records (EHR), Internet of Medical Things (IoMT), Multimodal Data Fusion, Tabular Transformer, Explainable AI.

I. INTRODUCTION

The integration of the Internet of Medical Things (IoMT) in hospitals, particularly in emergency departments, offers new opportunities for accelerating automated triage and improving early disease detection [1]. With this kind of environment, diagnostic decisions often rely on the combination of heterogeneous information. Those include vital signs, clinical observations, patient history, laboratory biomarkers, and specialty-specific examinations. To train robust machine learning models for this type of decision support, large amounts of multimodal medical data recreating real-world settings are required.

However, access to real-world patient data remains highly restricted because of privacy and legal constraints. Regulations such as GDPR and HIPAA impose strict limitations on the collection, storage, and secondary use of patient records. As a result, researchers frequently lack sufficiently large and diverse datasets to develop, test, and compare multimodal Deep Learning architectures in healthcare.

Synthetic data generation has emerged as a possible solution to this limitation. Unlike real-world medical data, synthetic records are not linked to actual patients and can therefore be

used for experimentation while reducing privacy concerns. In addition, synthetic data can be generated at scale and under controlled conditions, making it possible to simulate rare but clinically important pathologies that are often underrepresented in real datasets.

This study aims to address the structural weaknesses of purely data-driven generative models, which often output physiologically impossible values, by proposing a highly controlled expert-driven simulation engine. The main contributions of this work are threefold:

- **Multimodal Synthetic Generation:** We engineered a transparent, rule-based physiological pipeline utilizing truncated normal distributions to generate 500,000 synthetic patient records across 24 diagnostic classes, mathematically enforcing clinical coherence.
- **Real-World Distributional Validation:** We validated the clinical representativeness of our synthetic engine by conducting a distributional comparison against a real-world public Emergency Department dataset, ensuring epidemiological realism.
- **Comprehensive Benchmarking & XAI Validation:** We benchmarked Deep Learning models (Early Fusion MLP and Tabular Transformer) against an established tabular baseline (XGBoost), and used SHAP to verify whether the trained models rely on clinically meaningful features consistent with the generation rules.

The remainder of this paper is organized as follows. Section II presents related work, Section III describes the proposed method, Section IV reports the experimental results, and Section V concludes the paper.

II. RELATED WORK

Overcoming the medical data silo has driven significant research into synthetic generation. Deep Generative models, such as medGAN [2] and Variational Autoencoders (VAEs), are frequently used to synthesize tabular EHRs. While statistically powerful, these purely data-driven models often struggle to maintain logical and physiological constraints (e.g., preventing contradictory laboratory values). In contrast, our approach utilizes a rule-based, expert-driven physiological generation pipeline, supporting clinical coherence across multiple data modalities, making it suitable for validating diagnostic algorithms. However, rule-based engines carry the opposite risk: without empirical grounding, the generated distributions

may not reflect real clinical populations. Public Emergency Department resources, such as MIMIC-IV-ED and open-access hospital admission datasets, are increasingly used as reference distributions to quantify this gap, a practice we adopt to validate our generator.

Recent literature in medical AI emphasizes the necessity of multimodal fusion—synthesizing high-frequency sensor data with low-frequency clinical information [3]. While most IoMT frameworks focus on localized time-series analysis (e.g., detecting arrhythmias) [4], our synthetic dataset simulates the spatial fusion of these modalities into a unified tabular snapshot, representing the exact moment of emergency triage.

Tabular data remains a challenging domain for Deep Learning, historically dominated by tree-based methods. Gradient-boosted decision trees, most notably XGBoost, remain a strong baseline on structured clinical data due to their native handling of heterogeneous, sparse feature spaces. More recently, architectures like the FT-Transformer [5] have shown that projecting tabular features into embedding tokens allows models to capture complex, non-linear interactions via Self-Attention. Because neither family of models dominates uniformly across tabular benchmarks, we compare both paradigms in this work rather than restricting the evaluation to deep architectures alone. Furthermore, to ensure trust in these models, SHAP [6] has become a standard for Explainable AI (XAI), providing local and global feature importance that we use to validate the physiological integrity of our synthetic data across both model families.

A pervasive issue in medical Deep Learning is the class imbalance inherent to real-world epidemiological data, where rare, life-threatening conditions are outnumbered by benign cases. Traditional resampling techniques, such as SMOTE [7], often fail in multimodal contexts because they interpolate in the feature space without respecting physiological boundaries, generating clinically impossible synthetic samples. Our approach circumvents this limitation. By leveraging an expert-driven rule-based engine, we directly generate the minority classes (e.g., severe sepsis or hemorrhagic strokes) at scale. Combined with cost-sensitive learning algorithms, such as our weighted Cross-Entropy implementation, this ensures that the neural networks learn accurate, discriminative boundaries for critical conditions.

III. PROPOSED APPROACH

To address the challenges of multimodal IoMT data fusion, the limited availability of accessible medical datasets, and the need for empirically-grounded synthetic data, we designed an end-to-end framework combining a rule-based synthetic data generation algorithm, an external distributional validation step, data preprocessing, three benchmarked classification models, and a cost-sensitive training strategy.

The first component of our framework is a generation algorithm designed to simulate an emergency triage environment. Instead of learning patient distributions directly from real EHRs, the generator explicitly encodes physiological relationships to preserve medical coherence while avoiding

privacy constraints. Continuous variables are drawn from truncated normal distributions rather than clipped normal distributions, which prevents the artificial probability mass accumulation at physiological boundaries that a naive `clip()` operation would otherwise introduce.

The generation follows a two-level pipeline. Level 1 creates the initial triage profile using baseline demographic, vital-sign (e.g., HR, Blood Pressure, SpO₂), and observational variables (e.g., chest pain, confusion). Based on these indicators, patients are routed to Level 2 for specialty-specific examinations across Cardiology, Pulmonology, Neurology, and Infectious Diseases. Diagnosis-specific rules are grounded in clinically plausible thresholds; for instance, Troponin levels are sampled around 1800 ng/L for STEMI versus 250 ng/L for NSTEMI, BNP is escalated above 300 pg/mL for Acute Heart Failure, D-Dimers exceed 500 ng/mL under suspicion of Pulmonary Embolism, and Lactates surpass 2.5 mmol/L to characterize Severe Sepsis. This mimics the principle of multimodal healthcare fusion, where heterogeneous data sources are combined to improve diagnostics.

The pipeline covers 24 diagnoses, ranging from time-critical conditions (e.g., STEMI, severe sepsis) to benign overlapping conditions (e.g., anxiety hyperventilation) to introduce clinical ambiguity. To reproduce the sparsity of real-world hospitals, only clinically relevant specialty exams are generated. Missing tests therefore reflect clinical decisions rather than random errors. The output is a sparse multimodal tabular dataset of 500,000 synthetic patient records, approximating the heterogeneous information processed in IoMT-assisted triage systems.

A purely rule-based generator carries the risk of producing data that is internally coherent but not representative of real clinical populations. To mitigate this risk and to provide empirical evidence of clinical realism, we compared the marginal distributions of our synthetic vital signs against an independent, publicly available, real-world reference: an open-access dataset of 143,582 Emergency Department admissions from a large teaching hospital in Isfahan, Iran. Seven shared variables (Age, Heart Rate, Systolic and Diastolic Blood Pressure, Respiratory Rate, Temperature, and SpO₂) were aligned by name and unit, and their distributions were compared using kernel density estimation together with a two-sample Kolmogorov-Smirnov (KS) test, which quantifies the maximum distance between the empirical cumulative distribution functions of the synthetic and real samples. This procedure provides a transparent, reproducible, and quantified account of how close the generator is from a clinical distribution.

Diagnostic labels are merged into a single global target. Non-performed medical examinations, which previously were imputed with zeros, are now imputed using the per-feature median computed over patients for whom the test was actually performed. This change directly addresses a methodological risk identified during peer review: zero-imputation conflates two distinct pieces of information, namely “the test was not clinically indicated” and “the test returned a null value,” which could allow a classifier to shortcut the diagnostic task by learning test-ordering patterns rather than genuine physiolog-

ical characteristics. Median imputation limits this artificial signal by making a missing value statistically indistinguishable from a typical value of that biomarker. Continuous variables are subsequently standardized to ensure uniform gradient propagation for the neural architectures.

Medical datasets naturally exhibit strong class imbalance. Since classical interpolation techniques risk generating physiologically inconsistent samples when applied to heterogeneous medical variables, our approach avoids post-hoc oversampling. Instead, the generation engine directly controls the occurrence of minority diagnoses. In addition, imbalance is handled during training through a cost-sensitive learning strategy. We utilize a weighted Cross-Entropy loss function, defined for a single patient across C classes as:

$$\mathcal{L} = - \sum_{i=1}^C w_i y_i \log(\hat{y}_i) \quad (1)$$

where y_i is the binary indicator of the true class, $\hat{y}_i$ is the predicted probability for class i , and w_i is the weight assigned to class i . Minority classes receive higher weights, forcing the model to penalize errors on rare, life-threatening conditions. For the tree-based baseline, the same class weights are applied as per-sample weights during training.

The synthetic dataset is used to benchmark three classification models spanning both the deep learning and tree-based tabular paradigms, addressing the recommendation to compare against established machine learning methods.

Baseline MLP (Early Fusion): A fully connected network composed of two hidden layers (256 and 128 neurons), representing a classical early-fusion strategy. All modalities are concatenated into a single feature vector before processing. Batch Normalization and Dropout [8] are applied to mitigate overfitting.

Tabular Transformer (Contextual Fusion): A Transformer-inspired model adapted to tabular data. Each input feature is projected into a $d_{model} = 64$ embedding space. A learnable classification token is added, and the sequence is processed by Transformer encoders using the Scaled Dot-Product Attention mechanism, formulated as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (2)$$

where Q , K , and V represent the query, key, and value matrices, and d_k is the dimension of the key vectors. This contextual fusion allows the model to dynamically adjust the importance of each medical variable based on the patient's context.

XGBoost (Tree-Based Baseline): We include a gradient-boosted decision tree ensemble as a third, non-neural baseline. Trees natively partition the feature space along individual biomarker thresholds, which makes them a strong reference model for rule-generated data such as ours. Class imbalance is handled through per-sample weighting rather than an architectural mechanism.

Both neural models are trained using the same experimental protocol with an 80/20 train-test split. We utilize the AdamW

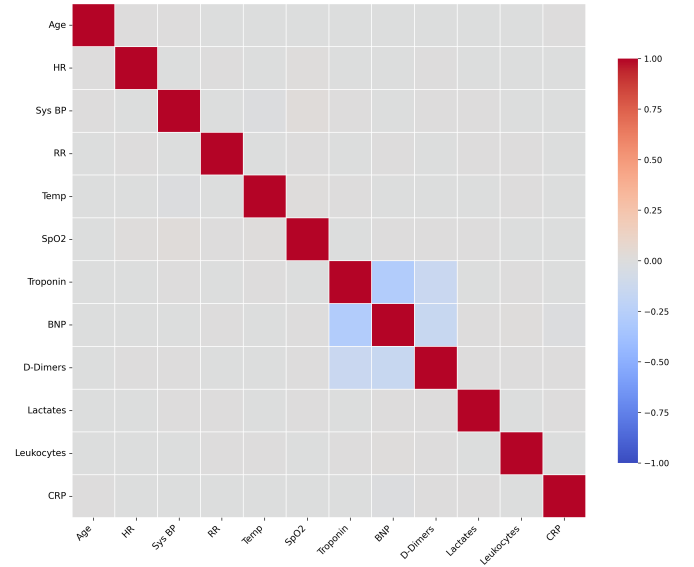


Fig. 1. Physiological correlation matrix over representative vitals and biomarkers, confirming internally coherent generation rules.

optimizer [9] (learning rate of 0.002, weight decay of 1×10^{-4}) and a batch size of 1024 to accelerate training. To ensure convergence, a ReduceLROnPlateau scheduler and an Early Stopping mechanism are implemented, restoring the best model weights at the end of training to minimize overfitting. The final evaluation encompasses Accuracy, Precision, Recall, Macro F1-score, ROC-AUC, Precision-Recall curves, computational cost, and SHAP-based interpretability for all three models.

IV. EXPERIMENTAL RESULTS

The evaluation of the proposed multimodal fusion framework was conducted on a 20% hold-out test set derived from the 500,000 synthetic patient records, and assesses three complementary aspects: the physiological coherence of the generated data, the predictive performance of the three benchmarked models, and the interpretability of the learned decision rules.

A rule-based generator can produce internally consistent data that nonetheless drifts from real clinical populations, so two independent checks are used before evaluating the classifiers. First, Fig. 1 reports the strongest absolute correlations among representative vital signs and biomarkers, confirming that dependencies emerge where physiologically expected (e.g., between inflammatory markers such as CRP and Leukocytes, and between cardiovascular indicators such as Troponin and BNP), rather than as random statistical noise.

Second, and more critically, we compare seven shared vital-sign variables against 143,582 real Emergency Department admissions from an independent, publicly available Iranian hospital dataset, using kernel density estimation and a two-sample Kolmogorov-Smirnov (KS) test. Table I and Fig. 2 summarize the results.

All differences are statistically significant ($p < 0.001$), which is expected given the sample sizes involved and is not

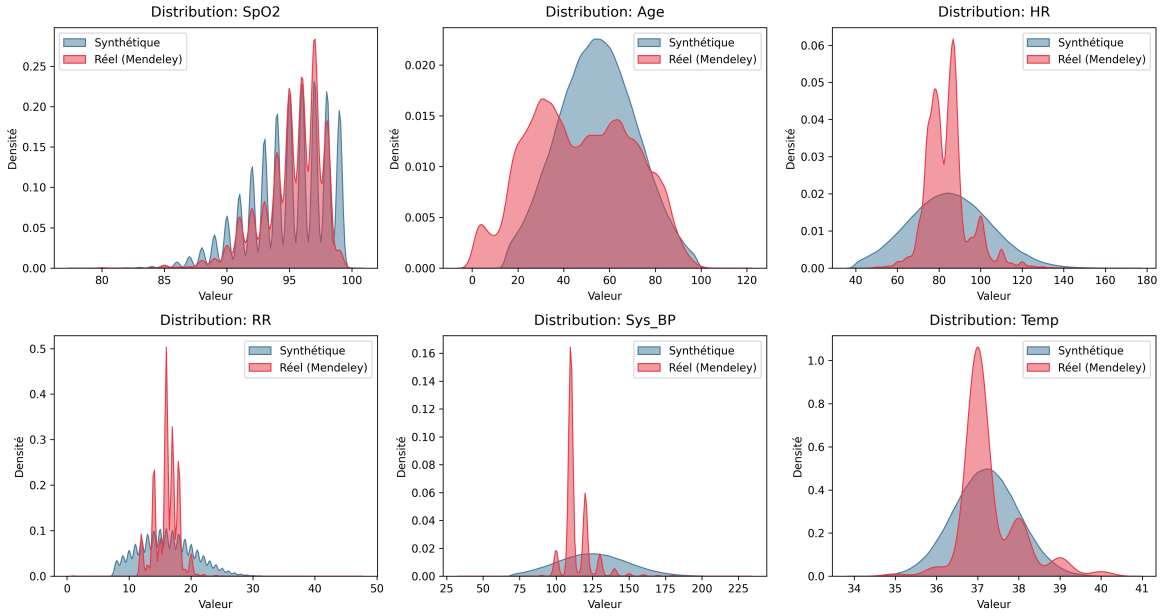


Fig. 2. Kernel density comparison between synthetic and real-world Emergency Department vitals, with Kolmogorov-Smirnov statistics (D) annotated per variable (Diastolic BP is omitted for layout compactness but quantified in Table I).

TABLE I
SYNTHETIC VS. REAL-WORLD DISTRIBUTIONAL COMPARISON

Variable	Synthetic (Mean $\pm$ SD)	Real (Mean $\pm$ SD)	KS D
SpO2	94.94 $\pm$ 2.98	95.23 $\pm$ 2.39	0.093
HR	85.18 $\pm$ 19.27	84.06 $\pm$ 10.30	0.227
RR	16.09 $\pm$ 4.47	16.08 $\pm$ 1.94	0.241
Age	54.77 $\pm$ 16.81	47.63 $\pm$ 22.24	0.215
Dia. BP	79.64 $\pm$ 14.74	72.15 $\pm$ 6.90	0.450
Sys. BP	125.33 $\pm$ 23.94	114.32 $\pm$ 10.74	0.464
Temp*	37.20 $\pm$ 0.80	37.29 $\pm$ 0.72	0.341

*Real Temperature is recorded for only 139/143,582 admissions; this row is reported for completeness but is not used to draw conclusions.

itself informative; the KS statistic D, bounded in [0,1], is the meaningful quantity. Central tendencies are well reproduced for SpO2, Heart Rate, and Respiratory Rate ($D \leq 0.24$), indicating that the generator captures realistic average physiology. The largest deviations occur on blood pressure ($D \approx 0.45$ -0.46): the real-world curves in Fig. 2 show sharp, narrow peaks at round numbers (110, 120, 130 mmHg), a digit-preference artifact of manual triage measurement rather than a true physiological distribution, whereas the generator produces a smoother continuous spread with a wider variance. This comparison therefore does not indicate that the synthetic data is unrealistic, but rather that it models an idealized continuous physiology instead of the measurement and rounding behavior of human triage staff, a distinction we make explicit rather than obscure.

Table II summarizes the training configuration shared by the two neural models; the tree-based baseline uses the same weighted-sample strategy without requiring these hyperparameters.

Table III reports Accuracy, macro-averaged Precision, Recall,

TABLE II
MODEL TRAINING HYPERPARAMETERS

Hyperparameter	MLP Baseline	Transformer
Optimizer	AdamW	AdamW
Learning Rate	0.002	0.002
Weight Decay	1×10^{-4}	1×10^{-4}
Dropout Rate	0.3 / 0.2	0.2
Batch Size	1024	1024
Embedding Dim	N/A	64
Attention Heads	N/A	4

TABLE III
OVERALL CLASSIFICATION PERFORMANCE (TEST SET)

Metric	MLP	Transformer	XGBoost
Accuracy (%)	74.4	78.3	74.6
Precision (%)	65.2	64.5	65.9
Recall (%)	65.1	65.4	65.4
Macro F1 (%)	63.5	62.6	64.3
ROC-AUC (macro)	0.993	0.994	0.993
Train time (s)	40.3	264.3	15.4
Inference time (s)	0.180	0.628	0.548

F1-score, macro ROC-AUC, and computational cost for the three classifiers, and Fig. 3 shows the corresponding ROC and Precision-Recall curves.

Across all three models, ROC-AUC and Average Precision remain markedly higher than Accuracy and Macro F1: a macro ROC-AUC of 0.993-0.994 and an Average Precision above 0.89 coexist with Macro F1 scores between 62.6% and 64.3%. This gap is a structural consequence of the task rather than an inconsistency: with 24 diagnostic classes, these threshold-independent, macro-averaged metrics aggregate performance over $\binom{24}{2}$ pairwise class separations, most of

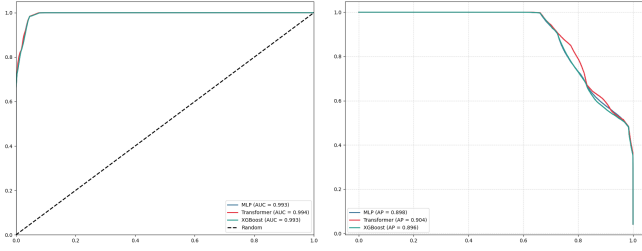


Fig. 3. ROC curves (left, AUC between 0.993 and 0.994) and Precision-Recall curves (right, Average Precision between 0.896 and 0.904) for the three benchmarked models.

which are easy (e.g., separating STEMI from Severe Migraine), which inflates the score. Accuracy and Macro F1, in contrast, are computed directly on the hardest decision boundaries at a fixed operating threshold and penalize every misclassification among clinically overlapping conditions equally, regardless of how rare the confusion is. No single model dominates across all four threshold-dependent metrics: the Transformer attains the highest Accuracy, Recall, and ROC-AUC, while XGBoost achieves the highest Precision and Macro F1. This divergence indicates that the Transformer classifies a larger share of majority-class patients correctly, whereas XGBoost is comparatively more conservative and precise on the harder, minority diagnoses that dominate the Macro F1 score; the Transformer’s Accuracy advantage over XGBoost (78.3% vs. 74.6%) is therefore not a uniform improvement but a shift in error distribution across classes.

Addressing previous recommendations to streamline visual analysis, the confusion matrices have been structurally compacted (Fig. 4) to facilitate direct side-by-side comparison. They further confirm that errors are not random: residual misclassifications concentrate on diagnosis pairs that share overlapping symptoms or biomarkers at triage time, such as Asthma Attack versus Anxiety-Induced Hyperventilation, or Severe Migraine versus Epileptic Seizure, rather than being spread uniformly across all 24 classes. This pattern indicates that the models struggle precisely where a human clinician would also require further examinations to disambiguate, which supports the dataset’s relevance as a challenging, realistic benchmark rather than a trivially separable one.

To verify that the classifiers rely on genuine physiological signal rather than spurious statistical artifacts, SHAP was applied independently to the neural network and to XGBoost (Fig. 5), since these two paradigms compute feature attributions through fundamentally different mechanisms (gradient-based versus tree-structure-based).

Comparing the top-8 most influential features of each model shows an overlap of 3/8 variables: Troponin, BNP, and Chest X-Ray finding are ranked among the most important drivers by both models, despite each model also relying on paradigm-specific secondary features (e.g., Creatinine and LVEF for the neural network; PCT, Lactates, and ECG ST-Elevation for XGBoost). This cross-architecture agreement on these clinical drivers is a stronger indicator of genuine

rule-learning than either explanation taken alone: it is unlikely that two structurally unrelated models would converge on the same variables by coincidence, so this shared core constitutes evidence that both models learned the intended physiological relationships embedded in the generation engine.

For real-world IoMT deployment, computational efficiency is as important as predictive accuracy, since edge and fog nodes operate under strict memory and energy budgets. As reported in Table III, the Tabular Transformer’s self-attention mechanism, which scales quadratically with the number of features, required 264.3 seconds to train against 40.3 seconds for the MLP and only 15.4 seconds for XGBoost, with inference time following the same trend (0.628 s versus 0.180 s and 0.548 s, respectively). The Transformer’s Accuracy and ROC-AUC advantage therefore comes at a training cost roughly 6 to 17 times higher than the other two models, for a Macro F1 that remains below XGBoost’s. Consequently, XGBoost offers the best overall accuracy-latency trade-off, followed closely by the MLP, making both substantially more viable than the Transformer for decentralized, edge-oriented IoMT deployment where retraining or on-device inference budgets are constrained.

V. CONCLUSION AND FUTURE WORK

This work addressed the limited availability of accessible medical datasets for Deep Learning in healthcare by proposing a rule-based synthetic data generation framework for multimodal IoMT-inspired triage, combining 500,000 synthetic patient records with an external empirical validation step and a three-model benchmark spanning both deep and tree-based tabular paradigms.

Two methodological risks identified during development were directly addressed rather than left unexamined. First, generating data from rules alone offers no guarantee of clinical realism; comparing our synthetic vitals against 143,582 real Emergency Department admissions showed that central tendencies (SpO2, Heart Rate, Respiratory Rate) are well reproduced, while blood pressure variance is underestimated because real triage measurements exhibit a digit-preference rounding pattern that a continuous generator does not reproduce – a concrete, quantified, and now explicit boundary of the framework rather than an untested assumption. Second, imputing missing laboratory tests with zero risk teaching the models to recognize test-ordering patterns instead of physiology; switching to median imputation removed this shortcut, and the resulting agreement between two structurally unrelated explainability methods (gradient-based SHAP on the neural network and tree-based SHAP on XGBoost) on the same top clinical drivers indicates that the classifiers rely on genuine physiological signal under the corrected preprocessing.

The benchmark itself confirms that the dataset is neither trivial nor degenerate: high macro ROC-AUC and Average Precision coexist with more moderate Accuracy and Macro F1 because residual errors concentrate on diagnosis pairs that are genuinely ambiguous at the point of triage, mirroring the kind of uncertainty a clinician would also face before further

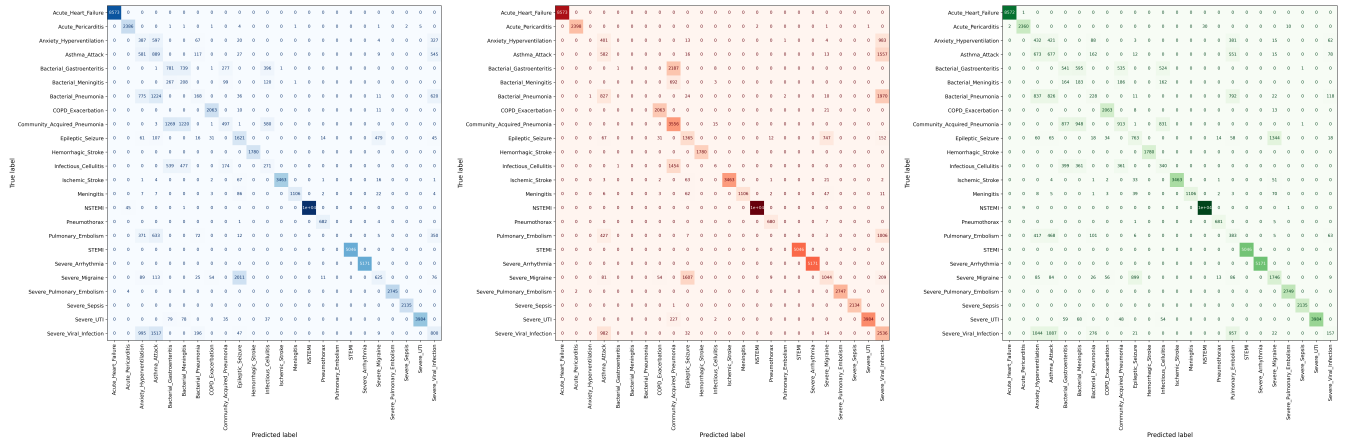


Fig. 4. Confusion matrices for the MLP, Tabular Transformer, and XGBoost classifiers, explicitly reduced in size for comparative layout, showing consistent confusion patterns concentrated among clinically overlapping diagnoses.

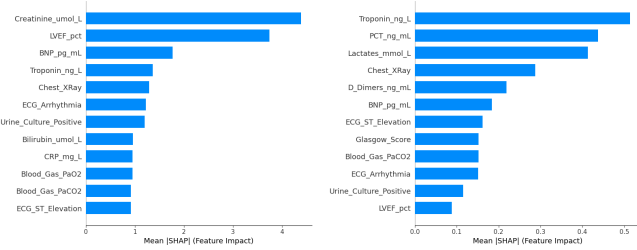


Fig. 5. Mean absolute SHAP feature importance for the Neural Network (left) and XGBoost (right), showing convergent reliance on clinically meaningful variables.

examination. Among the three models, no single architecture dominates every metric: the Transformer attains the highest Accuracy and ROC-AUC, while XGBoost achieves the highest Precision and Macro F1 at a training cost roughly an order of magnitude lower, confirming that a well-tuned tree-based baseline remains highly competitive with deep architectures on this structured tabular task; the MLP follows closely behind XGBoost on both accuracy and efficiency, whereas the Transformer’s quadratic attention cost makes it the least suitable of the three for resource-constrained, edge-oriented IoMT deployment despite its raw accuracy edge.

The framework nonetheless has clear limitations. The generated records remain structured tabular representations rather than including unstructured modalities such as radiological images or free-text notes; the Chest X-ray and Brain CT variables are encoded findings, not images. The dataset also represents a static triage snapshot rather than a longitudinal trajectory, and does not yet model the temporal evolution of vitals or laboratory values. Finally, external validation was limited to seven vital-sign variables shared with the reference dataset; laboratory biomarkers such as Troponin, BNP, and D-Dimers could not be cross-checked against a public source with matching fields, and the transferability of models trained purely on synthetic data to real clinical deployment remains to be established.

Future work will pursue three directions accordingly. First, validating laboratory biomarkers against a richer clinical resource, such as a credentialed extract of MIMIC-IV-ED, would close the remaining gap left by the vitals-only comparison performed here. Second, injecting measurement-realistic noise (e.g., digit-preference rounding) into the vital-sign generator could narrow the residual distributional gap observed on blood pressure. Third, extending the framework with longitudinal IoMT signals and unstructured modalities (synthetic imaging combined with tabular data through CNN or Vision Transformer fusion) would move it toward a more complete multimodal diagnostic benchmark for healthcare research.

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